

WhatsApp POS Receipt Integration Report

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Executive summary

The WhatsApp receipt slice has been implemented for the POS sales/payment flow as a controlled, enterprise-grade delivery option. Cashiers can choose WhatsApp as a receipt channel, enter a customer WhatsApp number, and must confirm explicit customer consent before a sale can be charged with WhatsApp receipt delivery.

Live WhatsApp sends are disabled by default. When enabled and configured, the integration sends through the WhatsApp Cloud API template boundary. When disabled or incomplete, the receipt delivery attempt is recorded as pending without blocking the sale.

Scope implemented

- Added WhatsApp as an explicit POS receipt channel in the shared receipt channel contract.
- Added required WhatsApp destination and explicit consent validation to POS receipt schemas.
- Added a WhatsApp receipt provider with phone normalization, destination redaction, destination hashing, safe config reporting, provider result mapping, and disabled-by-default behavior.
- Wired POS sale commit receipt delivery to forward WhatsApp consent into the receipt service.
- Wired the POS UI to show a WhatsApp destination field and consent checkbox only when WhatsApp is selected.
- Added English and French POS receipt copy for WhatsApp channel, destination, and consent states.
- Added tests for channel contract, consent validation, provider behavior, POS sale delivery resilience, legal delivery blocking, and UI state.

Security and control posture

- Customer consent is required before WhatsApp receipt delivery can be submitted.
- WhatsApp live sends require `STOQUIFY_WHATSAPP_RECEIPT_LIVE_SENDS=1` plus the required provider configuration.
- Raw WhatsApp destinations are normalized only at the provider boundary.
- Delivery results return redacted destinations and a hash, not the raw phone number.
- Audit metadata records redacted destination and destination hash.
- Provider configuration snapshots expose presence booleans and missing keys, not token values.
- Receipt delivery still respects legal/fiscal receipt delivery blocks.
- POS sale completion is not invalidated by provider unavailability; failed receipt delivery is surfaced as delivery status.

Files changed in this pass

- `.env.example`
- `components/pos/ProfessionalPOSSystem.tsx`

- components/pos/__tests__/ProfessionalPOSSystem.shift-close.test.tsx
- messages/en.json
- messages/fr.json
- services/communication/whatsapp-receipt.provider.ts
- services/communication/__tests__/whatsapp-receipt.provider.test.ts
- services/pos/receipt-channels.ts
- services/pos/pos.schemas.ts
- services/pos/pos.service.ts
- services/pos/receipt.service.ts
- services/pos/__tests__/receipt-channel-contract.test.ts
- services/pos/__tests__/receipt-public.test.ts
- services/pos/__tests__/pos.service.test.ts

Verification results

- node -e "...JSON.parse..." for messages/en.json and messages/fr.json: passed.
- npm run typecheck: passed.
- Focused lint over touched runtime and test files: passed.
- npm test -- --runInBand
services/communication/__tests__/whatsapp-receipt.provider.test.ts
services/pos/__tests__/receipt-channel-contract.test.ts
services/pos/__tests__/receipt-public.test.ts
services/pos/__tests__/pos.service.test.ts
components/pos/__tests__/ProfessionalPOSSystem.shift-close.test.tsx: passed, 5 suites, 41 tests.
- npm test -- --runInBand services/pos/__tests__ actions/pos/__tests__: passed, 17 suites passed, 1 skipped, 134 tests passed, 7 skipped.

Policy gate result

npm run policy: gates was attempted and stopped at inventory: boundary: fail.

The failure is not from the WhatsApp receipt implementation. The gate is currently scanning generated Next build folders, especially .next-dev/ and .next-slice410/, and reporting 66 direct stock mutation findings inside generated server output. The exact remaining failing command is:

```
node scripts/inventory-boundary-gate.js --mode fail
```

Recommended follow-up: adjust the inventory boundary gate to exclude generated build output folders such as .next-dev/, .next-slice410/, and other .next* artifacts, or remove stale generated build directories before policy gates run.

Remaining notes

- WhatsApp Cloud API template names, language, phone number ID, and access token must be supplied by environment only.
- No WhatsApp token or raw customer phone number is intentionally written to logs, audit metadata, or provider results.
- This slice is limited to POS receipt delivery and is structured so later WhatsApp workflows can reuse the provider boundary without making WhatsApp the default communication layer.